This assignment will not be graded and is only for practice.

Limits:

Level 1:

1) Find the value of the following limit:

$$\lim_{(x,y)\to(2,3)} \frac{x^2+y^2-y}{y^2-x^2}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 2

1 point

2) Suppose f and g are multivariable functions defined on domain $\mathbb{R}^2$ with the following limits: ***1 point***

$$\lim_{(x,y)\to(0,0)} f(x,y) = l \text{ and } \lim_{(x,y)\to(0,0)} g(x,y) = k$$

where $l, k \in \mathbb{R}$. Which of the following options is(are) correct?

- [ ] $\lim\limits_{(x,y)\to(0,0)} (f(x,y) + g(x,y)) = l + k$
- [ ] $\lim\limits_{(x,y)\to(0,0)} (f(x,y).g(x,y)) = lk$
- [ ] $\lim\limits_{(x,y)\to(0,0)} (5f(x,y)) = 5l$
- [ ] $\lim\limits_{(x,y)\to(0,0)} (\frac{f(x,y)}{g(x,y)}) = \frac{l}{k}$, when $k = 0$

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\lim\limits_{(x,y)\to(0,0)} (f(x,y) + g(x,y)) = l + k$

$\lim\limits_{(x,y)\to(0,0)} (f(x,y).g(x,y)) = lk$

$\lim\limits_{(x,y)\to(0,0)} (\frac{f(x,y)}{g(x,y)}) = \frac{l}{k}$, when $k = 0$

3)

Consider the function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined as: $f(x,y) = \begin{cases} \frac{4x^2y^2}{x^2+y^2} & x,y = 0 \\ 0 & x = y = 0 \end{cases}$ What is the value of $\lim\limits_{(x,y)\to(0,0)} (f(x,y))$?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

4) Suppose f and g are multivariable functions defined on domain $\mathbb{R}^2$ as $f(x,y) = \frac{x^2+y^2+y(y-2x-1)}{x^2-y^2}$ and $g(x,y) = \frac{2y^2+x(x-2y-1)}{x^2-y^2}$. Find the value of $\lim\limits_{(x,y)\to(1,1)} (f(x,y) - g(x,y))$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0.5

1 point

5) Find the value of the following limit:

$$\lim_{(x,y)\to(3,1)} \frac{x^2-3xy^2}{x^2-9y^4}$$

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0.5

1 point

Level 2:

· Consider the function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined as:

$$f(x,y) = \begin{cases} \frac{3xy}{x^2+y^2} & x,y = 0 \\ 0 & x = y = 0 \end{cases}$$

Use the given information to answer the questions 6-8:

6) If f approaches to L as (x, y) approaches to the origin along Y-axis, then find the value of L ?

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

7) If f approaches to L as (x, y) approaches to the origin along a straight line $y = 3x$, then find the value of L

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0.9

1 point

8) Which of the following options is(are) false? **1 point**

- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$
- [ ] $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 1$
- [ ] Limit of the given function at (0,0) exists but is not equal to 0.
- [ ] Limit of the given function at (0,0) does not exist.

No, the answer is incorrect.
Score: 0

Accepted Answers:

$\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$
$\lim\limits_{(x,y)\to(0,0)} f(x,y) = 1$
Limit of the given function at (0,0) exists but is not equal to 0.

9) Which of the following functions have limit value equal to 5 at the point (1,1,-1)? **1 point**

- [ ] $f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2}$
- [ ] $f(x,y,z) = \frac{3(x+y-z)+7x^2y+yz}{xyz+zy+xz}$
- [ ] $f(x,y,z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx$
- [ ] $f(x,y,z) = \frac{x^2e^{\frac{x}{2}}+y^2e^{\frac{y}{2}}+z^2e^{\frac{-z}{2}}}{e^{\frac{x+y+z}{2}}}$

No, the answer is incorrect.
Score: 0

Accepted Answers:

$f(x,y,z) = \frac{x+3yz^2-zx}{xy+yz+xz^2}$
$f(x,y,z) = x^3 + y^3 + z^3 - 2xy - 2yz - 4zx$

10) Consider the function $f : \mathbb{R}^2 \longrightarrow \mathbb{R}$ defined as: **1 point**

$$f(x,y) = \begin{cases} \frac{x^k y}{x^{2n}+y^{2n}} & x,y = 0 \\ 0 & x = y = 0 \end{cases}$$

where $k, n \in \mathbb{N} \setminus \{0\}$. Which of the following statements is(are) true about f?

- [ ] If $k = 2n - 1$, then the limit at (0,0) exists and is equal to 0.
- [ ] If $k < 2n - 1$, then the limit at (0,0) does not exist.
- [ ] If $k > 2n$, then the limit at (0,0) always exists and is equal to 0.
- [ ] If $k > 2n$, then the limit at (0,0) does not exist.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $k < 2n - 1$, then the limit at (0,0) does not exist.
If $k > 2n$, then the limit at (0,0) always exists and is equal to 0.

Your score is: 0/10